Appendix · Multi-position decode probes (Experiment A)

Setup

- Captured residual at decode positions 0..15 of "The answer is: <num>".
- 1000 SVAMP problems × 16 positions × 13 layers × 768-dim.
- Per (position × layer): logistic-regression probe with 5-fold CV for

units digit, tens digit, and operator class.

Token at each position (most-common across 1000 examples)

- pos 0..3: 'The', 'answer', 'is', ':' (template, identical across all 1000)
- pos 4: FIRST variable token (model emits the answer's first digit)
- pos 5: EOS for 1-digit answers (831/1000) or second digit for multi-digit
- pos 6+: trailing decay

Headline finding (peak accuracy across 13 layers)

- pos 0 ('The'): units 18.9% tens 38.8% op 89.1%
- pos 4 (digit): units 39.6% tens 45.5% op 91.0%
- pos 5 (digit/eos):units 41.6% tens 48.9% op 91.0%
- pos 15 (decay): units 27.4% tens 40.0% op 72.3%

Reading

- Operator commitment is fully decided BEFORE decoding starts (89% at pos 0,

~91% across the answer positions, slow decay afterward).

- Even at pos 4 — where the actual answer digit is being emitted — units

accuracy is only ~40% and tens ~45%. The model encodes magnitude/operator,

not exact digits.

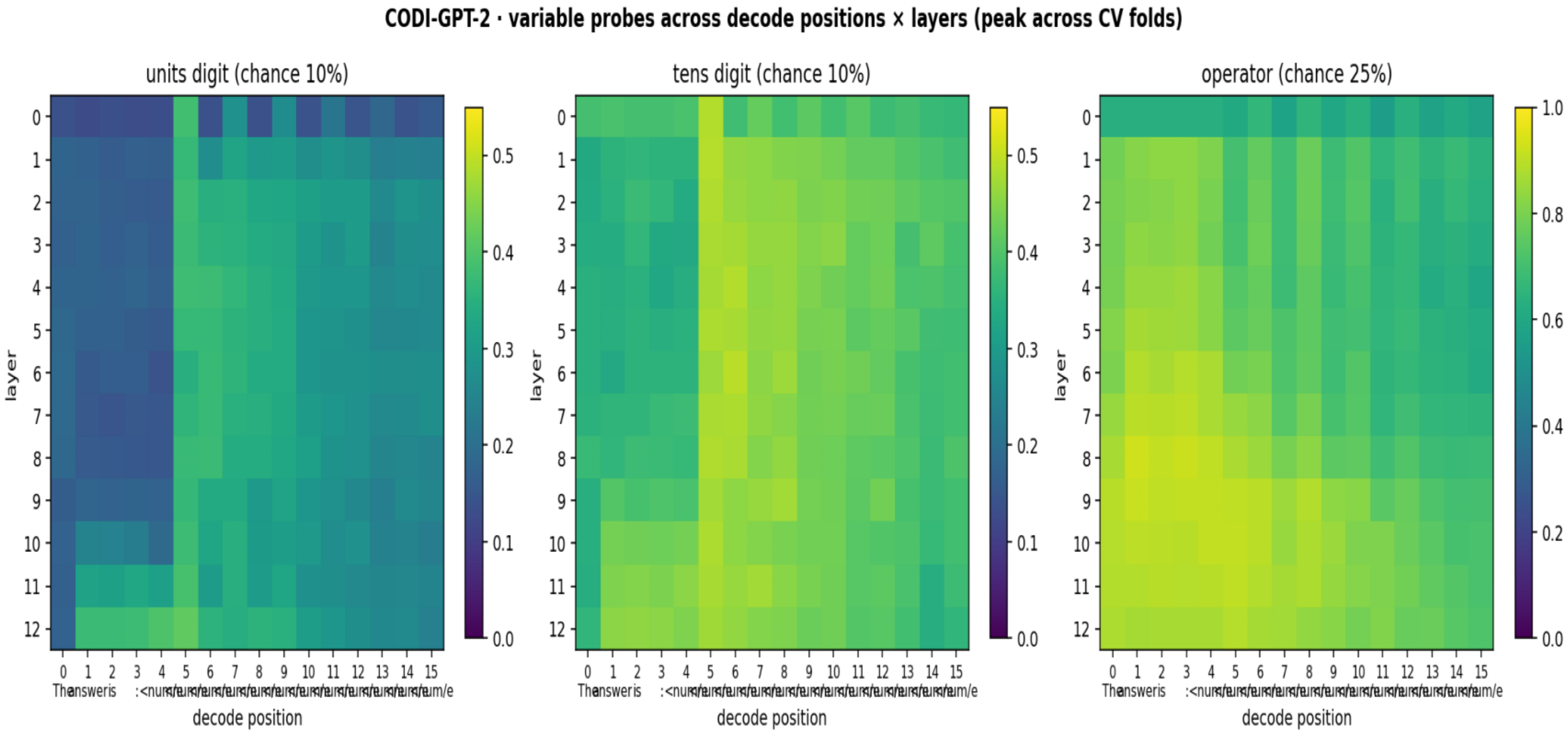
- The ~2 pp jump from pos 0->1 (units 19% -> 37%) shows the residual already

carries digit info during the template — the loop has happened.

Conclusion

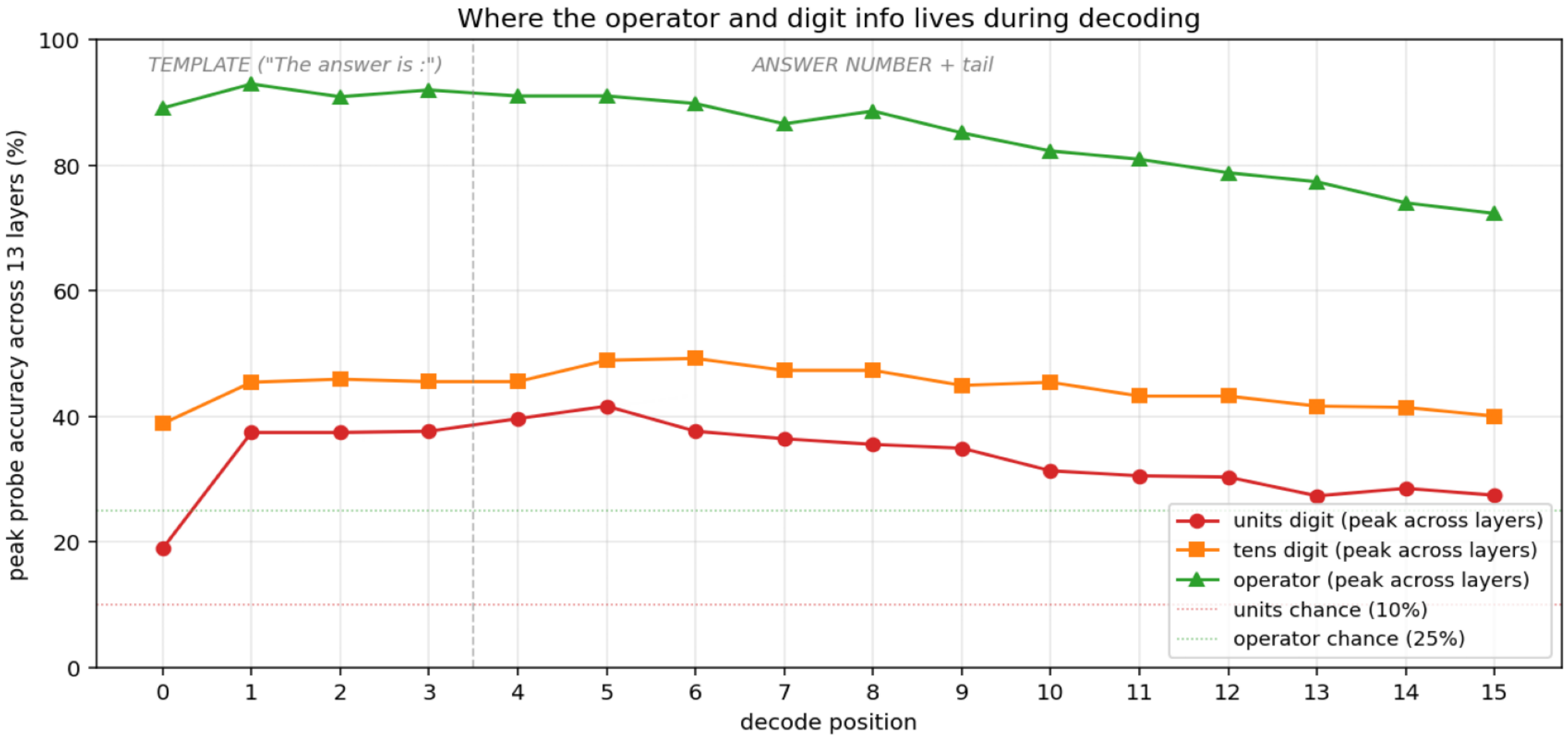
- The 'addition' is not a clean digit-level computation. Operator is decided

Per-(position × layer) probe accuracy heatmaps



Operator (right) is bright across all positions/layers — strongly encoded everywhere. Units (left) and tens (middle) are dim outside the answer-position band, and even there only reach ~40-50%.

Peak probe accuracy per position



) plateaus at ~89-93% from pos 0-7, then decays. Tens (orange) peaks ~49% at pos 6. Units (red) peaks ~42% at pos 5. Digit info is always near-floor, never the clean step-up you'd expect if the model were doing